JURUTERA SIBER RAWSEC

University Research Collaboration Portal

JURUS Level 1 (Analyst) Cyber Engineering Challenge

Role:	Lead Cyber Engineering Analyst	**Duration:**	Three (3) Weeks
Focus:	Academic Research Portal Infrastructure	**Level:**	Level 1 Competency

BACKGROUND

A public university consortium has launched a national research collaboration initiative aimed at improving cooperation between academic institutions, researchers, postgraduate students, and external industry partners.

As part of this initiative, the consortium plans to deploy a Research Collaboration Portal that enables users to upload research proposals, share project documentation, manage collaboration requests, publish announcements, and coordinate research activities through a centralised web-based platform.

During a recent infrastructure assessment, the consortium identified that its existing systems are outdated, inconsistently managed, and lack sufficient security controls, operational visibility, and recovery capabilities necessary to support a modern collaborative environment exposed to external users.

The consortium therefore requires a secure and operational infrastructure capable of supporting the new platform while maintaining service reliability, secure user access, accountability of administrative actions, and recoverability during operational incidents.

Your organisation has been engaged as a cyber engineering consultancy to design and implement the required solution.

You have been assigned as the lead Cyber Engineering Analyst responsible for delivering a production-ready environment.

Engineering-First Philosophy

Unlike traditional cybersecurity competitions, no infrastructure, software platforms, deployment templates, reference architectures, or predefined technologies will be provided.

Participants are expected to independently determine appropriate operating systems, deployment approaches, security mechanisms, monitoring solutions, operational controls, and backup strategies using practical engineering judgement and industry-relevant practices.

The objective of this challenge is not to assess familiarity with any specific product or vendor, but rather to evaluate the participant's ability to independently design, deploy, secure, monitor, troubleshoot, document, and recover a system using sound cyber engineering principles.

COMPETITION DURATION

Participants will be given three (3) weeks to complete the challenge.

During this period, participants may build and deploy their solution using any suitable environment of their choosing.

This may include cloud infrastructure, local virtualisation platforms, containerised deployments, physical hardware, or hybrid environments.

All infrastructure provisioning, software installation, configuration activities, testing, troubleshooting, security implementation, documentation, and demonstration preparation preparation are the responsibility of the participant.

PROJECT REQUIREMENTS

The university consortium requires a publicly accessible web-based platform capable of supporting collaborative research activities and document management services.

The chosen solution must be deployed on an operating system selected by the participant and must incorporate appropriate security controls to protect the system, user accounts, uploaded content, and operational services.

The deployed environment must demonstrate secure administrative access and appropriate user management practices.

Participants are expected to implement controls that reduce the likelihood of unauthorised access while maintaining operational usability for legitimate administrators, researchers, and collaborators.

The solution should demonstrate evidence of secure baseline configuration practices and appropriate hardening measures suitable for production-facing infrastructure.

The application environment must include a database component capable of storing user account information, project records, uploaded document metadata, and collaboration requests.

Participants are expected to implement suitable database security controls and demonstrate an understanding of least-privilege principles, access management, secure credential handling, and secure service configuration practices.

The consortium has also identified a requirement for operational visibility and auditability.

Participants are therefore required to implement monitoring, logging, or observability capabilities that enable administrators to identify system activity, user actions, file access events, operational issues, and potentially security-relevant incidents.

The technologies selected to fulfil these requirements are entirely at the discretion of the participant.

In addition, the consortium requires assurance that critical services and research-related data can be restored in the event of accidental deletion, infrastructure failure, service disruption, or operational incidents.

Participants must therefore design and implement an appropriate backup and recovery approach capable of restoring business-critical functionality and information.

Participants are encouraged to prioritise maintainability, operational simplicity, secure collaboration practices, and practical recovery procedures throughout the implementation.

DELIVERABLES

At the conclusion of the challenge, participants shall submit a technical package containing sufficient evidence to demonstrate the design, implementation, security, operation, monitoring, and recoverability of their solution.

The submission should include:

- Solution architecture diagram
- Technical implementation documentation
- Security configuration evidence
- Administrative access management documentation
- Monitoring and logging evidence
- Backup and recovery procedures
- Operational validation or testing evidence
- Supporting artefacts relevant to the implementation

Evidence may include screenshots, configuration files, logs, scripts, command outputs, deployment notes, validation results, diagrams, or any additional material considered relevant by the participant.

Participants should assume that assessors have not previously seen their environment.

Documentation should therefore be sufficiently detailed to allow assessors to understand the participant's engineering decisions, deployment methodology, security considerations, and operational approach.

LIVE DEMONSTRATION

Each participant or team will be allocated thirty (30) minutes for a live demonstration and technical defence session.

During this session, participants will be expected to present their architecture and explain the rationale behind their deployment decisions, technology selections, security controls, monitoring implementation, and recovery strategy.

Participants must demonstrate that their solution is operational and capable of supporting the stated business requirements.

Assessors may request demonstrations involving:

- Administrative access controls
- User and permission management
- Database protection mechanisms
- File access or upload protections
- Monitoring and logging functionality
- Backup validation
- Recovery procedures
- Troubleshooting activities

The demonstration should also provide evidence that participants understand the engineering decisions they have made throughout the project lifecycle.

Assessors may ask participants to discuss alternative approaches, explain trade-offs, justify technology selections, and describe how operational or security risks were mitigated.

Participants should be prepared to defend both their technical implementation and the reasoning behind their engineering decisions.

ASSESSMENT PHILOSOPHY

The JURUS programme follows an engineering-first assessment philosophy.

Assessment will therefore focus on demonstrated capability rather than theoretical knowledge alone.

Participants will be evaluated based on:

- Quality and completeness of implementation
- Effectiveness of security controls
- Operational usability and maintainability
- Clarity and completeness of documentation
- Visibility provided through monitoring and logging

- Practicality of backup and recovery procedures
- Ability to explain and justify engineering decisions

A more complex solution will not necessarily receive a higher score. Greater emphasis will be placed on solutions that are appropriate, secure, maintainable, operationally practical, well-documented, and clearly understood by the participant.

The highest-scoring submissions will demonstrate not only technical competence, but also strong engineering judgement, structured operational thinking, and professional communication skills.

EXPECTED OUTCOME

Upon completion of the challenge, successful participants should demonstrate the capability to independently design, build, secure, monitor, troubleshoot, document, and recover a production-ready collaborative system using industry-relevant cyber engineering practices.

This outcome aligns directly with the JURUS Level 1 Analyst competency objective of developing practitioners capable of independently building and securing operational systems while applying structured troubleshooting methodologies, operational discipline, and sound engineering judgement.